Drew University - CSCI 260 - Fall 2024
In-Class Exercise 1 - Data Representation

Instruction: Please work in groups of 3 and write down your names for attendance. 09/11/2022
Names:

1. Given the following binary numbers, compute their decimal values when being treated as unsigned, sign magnitude, 1's complement, and 2's complement numbers. (Just give me the formula.)

   (a) $(0010010111)_2$

   (b) $(1110000110)_2$

2. Given the following decimal numbers, compute their 12-bit 2's complement binary representations and convert the results into their corresponding hexadecimal forms.

   (a) $(-56)_{10}$ using short division.

   (b) $(175)_{10}$ by subtracting powers of 2.

3. Compute the following logical expressions.

   (a) NOT(01010111 AND NOT(11010111))

   (b) 0011 OR (0110 AND 1101)

   (c) (0011 OR 0110) AND (0011 OR 1101)

4. For 2's complement binary numbers, compute the followings and determine whether there is overflow in each of the results:

   (a) $01100111 + 10110$:

   (b) $101111 + 101101$

   (c) $01100111 - 10110$: